Practical 2 — Numerical business data

Comparing cereal costs · 307312 Applied Statistics — Masters Level

- This practical accompanies the lessons on numerical variables, histograms, central tendency and dispersion in Part 2: Descriptive statistics.
- Complete Practical 1 first.
- Allow approximately **50–60 minutes**.

By the end, you should be able to:

- Inspect a small business dataset.
- Describe a distribution with a histogram.
- Calculate its centre and spread.
- Use quartiles and boxplots to investigate unusual values.
- Explain what the results can tell a manager.

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1. The business question

- A grocery buyer is reviewing a historical list of **33 breakfast cereals**.
- Before comparing individual products, the buyer wants a short description of their costs:

> What is a typical cost per ounce in this list, and how much do the costs vary?

You will prepare a histogram, a summary table and a short recommendation about how to describe these costs.

- The records come from the **CEREALS** dataset accompanying Berenson, Levine and Krehbiel's *Basic Business Statistics: Concepts and Applications*, ninth edition.
- The book attributes the data to *Consumer Reports*, October 1999.
- These are historical records, so the amounts describe this list rather than current shop prices.

2. Open and inspect the data

- Open `Applied-Statistics.Rproj` in RStudio, then open this notebook.
- Run the code cells in order.
- The CSV is stored in `data/practicals/cereals.csv`, relative to the course project folder.
- The import cell below locates this same folder when run from either the course root or `practicals/`; keep the data outside `practicals/`.
- You can also download the CSV.

- The only package used here is **ggplot2**, introduced in Part 1.
- If it is not installed, run `install.packages("ggplot2")` once in the Console.

```
# read.csv() imports the supplied observations. Product labels stay as text.
# Find the shared data folder from either the course root or this notebook folder.
data_dir <- if (dir.exists("data")) "data" else file.path("..", "data")
cereals <- read.csv(file.path(data_dir, "practicals/cereals.csv"),
         stringsAsFactors = FALSE)
source(file.path(data_dir, "..", "practicals", "_includes", "petra-plots.R"))

# nrow() counts observations; str() displays variable names and R data types.
nrow(cereals)
```

```
[1] 33
```

```
str(cereals)
```

```
'data.frame':   33 obs. of  5 variables:
 $ product       : chr  "Kellogg's" "Sam's Choice Extra raisin (Wal-Mart)" "Kountry
Fresh (Winn-Dixie)" "Post Premium" ...
 $ cost_per_ounce: num  0.13 0.1 0.1 0.14 0.11 0.12 0.15 0.11 0.19 0.17 ...
 $ calories      : num  200 210 170 190 170 200 200 200 180 200 ...
 $ fiber_g       : num  8 5 7 8 7 8 8 8 5 8 ...
 $ sugar_g       : num  18 23 17 20 17 18 18 18 19 1 ...
```

One row represents one cereal product entry. A row does not represent a customer, a purchase or a serving eaten by someone.

Variable	Meaning	How to treat it
`product`	Product label supplied with the data	Text label used to identify an entry
`cost_per_ounce`	Cost in dollars per ounce, as reported	Numerical measurement; our variable of interest
`calories`	Reported calories	Numerical; serving basis is not specified in this file
`fiber_g`	Reported fibre in grams	Numerical; serving basis is not specified in this file
`sugar_g`	Reported sugar in grams	Numerical; serving basis is not specified in this file

- We use cost **per ounce** because its comparison basis is explicit.
- We cannot assume that the nutrition figures refer to equal-sized servings or rank products as healthier from these columns alone.

```r
# head() selects the first six rows; kable() displays them as a readable table.
knitr::kable(head(cereals), digits = 2,
             col.names = c("Product", "Cost ($/oz)", "Calories", "Fibre (g)",
         "Sugar (g)"))
```

Product	Cost ($/oz)	Calories	Fibre (g)	Sugar (g)
Kellogg's	0.13	200	8	18
Sam's Choice Extra raisin (Wal-Mart)	0.10	210	5	23
Kountry Fresh (Winn-Dixie)	0.10	170	7	17
Post Premium	0.14	190	8	20
American Fare (kmart)	0.11	170	7	17
America's Choice (A&P)	0.12	200	8	18

- Check missing values before calculating summaries.
- A missing value and a recorded zero mean different things.

```r
# is.na() marks missing cells; colSums() counts them separately for each variable.
colSums(is.na(cereals))
```

```
      product cost_per_ounce      calories       fiber_g       sugar_g
            0             0             0             0             0
```

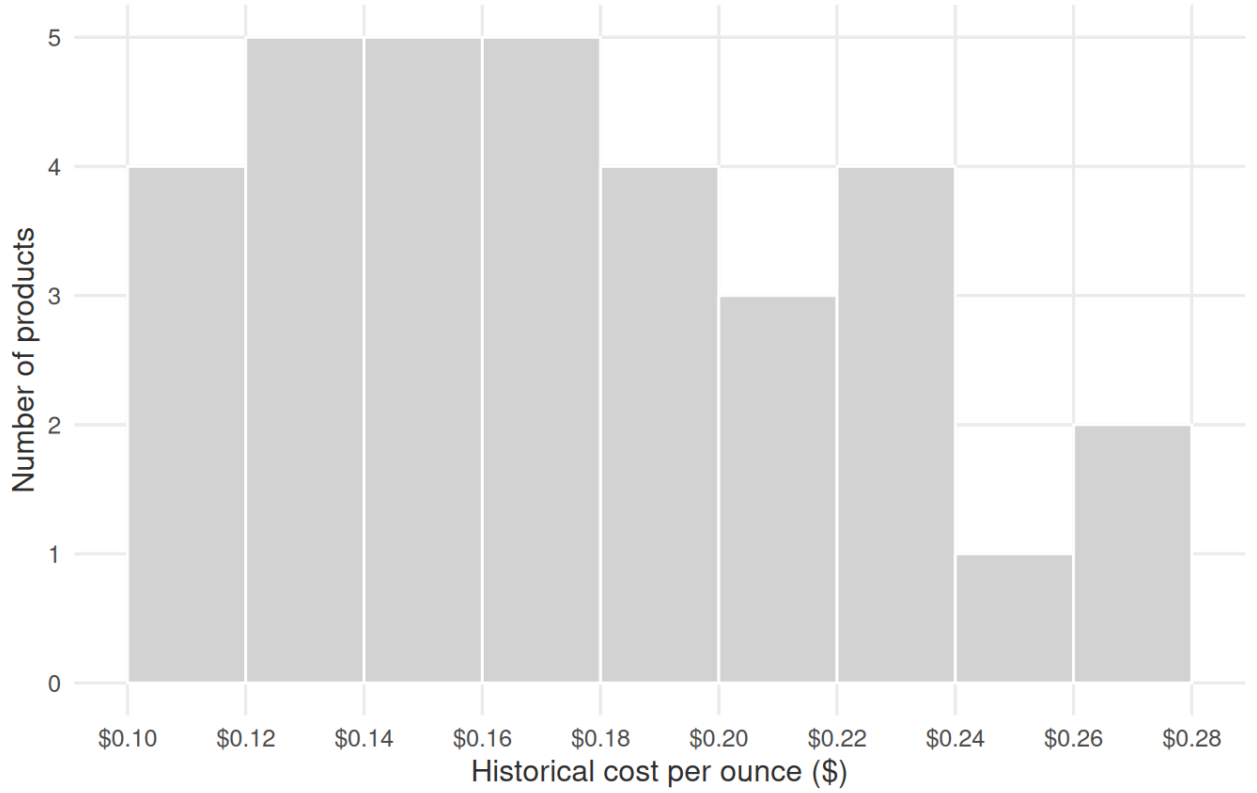
- There are no missing values in this supplied table.
- The zero sugar values are recorded observations; they are not missing-value codes.

Pause and explain: Why would taking the average of the `product` column make no sense, while taking the average of `cost_per_ounce` does?

3. Look at the distribution

- A histogram groups numerical costs into intervals.
- Its bars count **products**, rather than purchases or customers.
- Each interval below is $0.02 per ounce wide.

```r
# aes() maps cost to the horizontal axis; geom_histogram() counts products in
       bins.
# boundary and closed specify the bin edges and their inclusion convention.
ggplot(cereals, aes(x = cost_per_ounce)) +
  geom_histogram(binwidth = 0.02, boundary = 0.10, closed = "left",
                 fill = petra_accent, colour = "white") +
  scale_x_continuous(breaks = seq(0.10, 0.28, by = 0.02),
                     labels = function(x) sprintf("$%.2f", x)) +
  labs(x = "Historical cost per ounce ($)", y = "Number of products")
```



Historical costs of 33 cereal products. Each bin spans $0.02 per ounce.

- Before calculating anything, describe where most costs lie and how far the observed values extend.
- Look for gaps or unusually high or low costs.
- A product is not an error simply because its cost is unusual.
- These prices are rounded to cents, and changing the bin width can change the histogram's appearance.
- Avoid giving a strong shape label based on one arrangement of a small number of bars.

4. Describe a typical cost

- The **mean** uses every cost.
- The **median** is the middle cost after sorting the observations.
- Comparing them helps us assess whether the choice of a measure of centre makes much difference here.

```
# Save the numerical column so the calculations below are easy to read.
cost <- cereals$cost_per_ounce
mean_cost <- mean(cost)
median_cost <- median(cost)

centre <- data.frame(
  Measure = c("Mean", "Median"),
  Dollars_per_ounce = c(mean_cost, median_cost)
)
knitr::kable(centre, digits = 4,
             col.names = c("Measure", "Historical cost ($/oz)"))
```

Measure	Historical cost ($/oz)
Mean	0.1706
Median	0.1700

- Both measures are approximately **$0.17 per ounce**.
- For this list, either gives a similar description of the typical cost.
- The small difference between them is much less than one cent per ounce.
- This agreement does not prove that the distribution is symmetric or normal.
- We still need the plot and the spread.
- The additional decimal places in the calculated mean result from averaging prices; the original costs were reported to cents.

5. Describe the variation

- A typical value does not tell the buyer whether all products cost about the same amount.
- We also need to describe how much the costs differ.

```
# range() returns the minimum and maximum; diff() subtracts them.
# sd() uses the sample denominator n-1. IQR() measures the middle-half spread.
spread <- data.frame(
  Measure = c("Minimum", "Maximum", "Range", "Sample standard deviation",
        "Interquartile range"),
  Dollars_per_ounce = c(min(cost), max(cost), diff(range(cost)), sd(cost),
        IQR(cost))
)
knitr::kable(spread, digits = 4,
             col.names = c("Measure", "Historical cost ($/oz)"))
```

Measure	Historical cost ($/oz)
Minimum	0.1000
Maximum	0.2700
Range	0.1700
Sample standard deviation	0.0469
Interquartile range	0.0700

- **Minimum and maximum:** the listed costs run from $0.10 to $0.27 per ounce.
- **Range:** the highest and lowest costs differ by $0.17 per ounce. This uses only the two extremes.
- **Sample standard deviation:** approximately $0.047 per ounce, or 4.7 cents. It describes spread around the mean using squared deviations; it is not the average absolute difference from the mean.
- **Interquartile range:** $0.07 per ounce. Using R's default quartile convention, the middle half of the recorded costs runs from $0.13 to $0.20 per ounce.

- We use the sample-SD convention taught in Part 2.
- Calculating it does not establish that these products were randomly sampled from all cereals.

Pause and explain: Why is “the typical cost is about $0.17 per ounce” an incomplete description of the list?

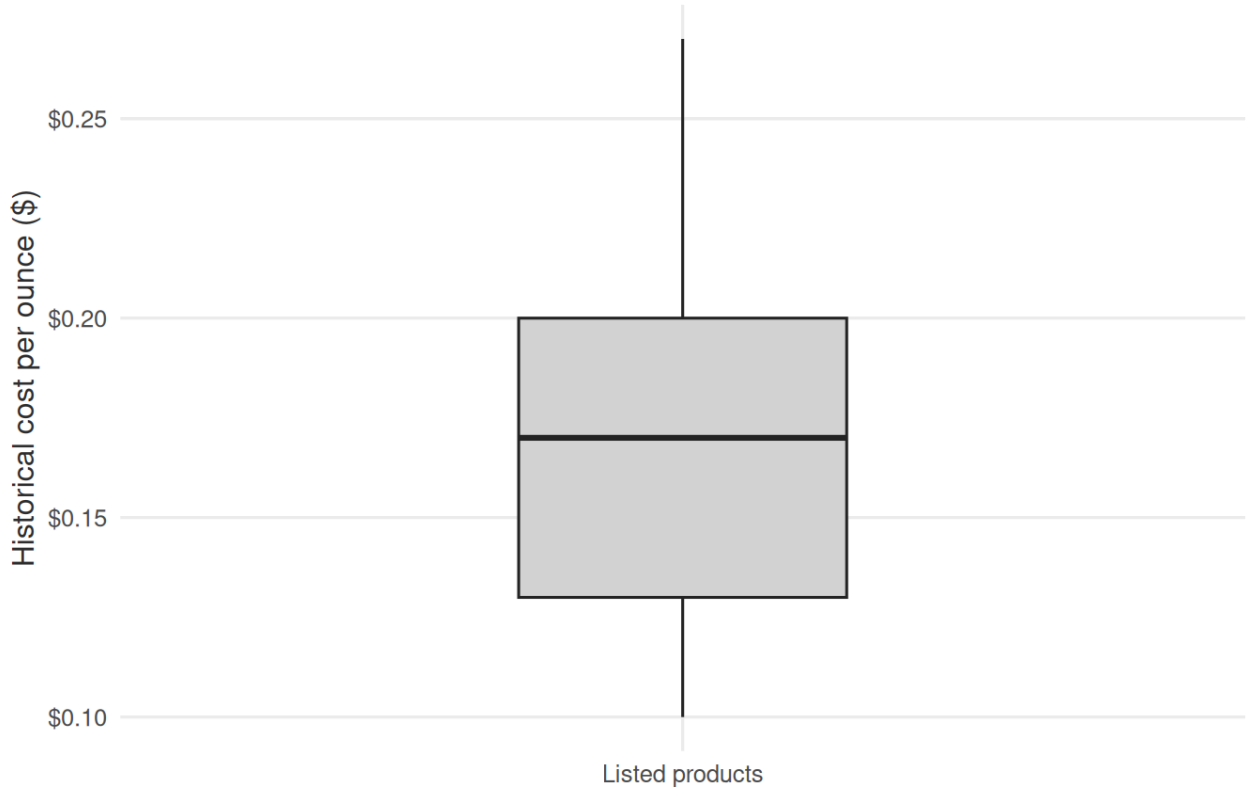
6. Quartiles, boxplots and unusual values

- The histogram and numerical summaries describe different aspects of the same observations.
- A boxplot adds a compact view of the middle half and flags values beyond its whiskers.

```
# R type-7 quantiles use interpolation; state the convention when comparing
        software.
quartiles <- quantile(cost, probs = c(0.25, 0.5, 0.75), type = 7)
knitr::kable(data.frame(Quartile = names(quartiles), Cost_per_ounce =
        unname(quartiles)),
             digits = 3)
```

Quartile	Cost_per_ounce
25%	0.13
50%	0.17
75%	0.20

```
# The boxplot convention can differ from type-7 quartiles for some sample sizes.
ggplot(cereals, aes(x = "Listed products", y = cost_per_ounce)) +
  geom_boxplot(fill = petra_accent, colour = "#262626", width = 0.35) +
  scale_y_continuous(labels = function(x) sprintf("$%.2f", x)) +
  labs(x = NULL, y = "Historical cost per ounce ($)")
```



Historical cereal costs. The box shows the middle half of the observations.

- Calculate 1.5-IQR fences using the stated type-7 quartiles.
- A fence need not be an observed value.
- A boxplot whisker ends at an observation within its plotting fences; it is not automatically the minimum or maximum in every dataset.

```
# Keep all records; the flag identifies values to investigate, not values to
        delete.
q <- quantile(cost, c(0.25, 0.75), type = 7)
iqr <- q[2] - q[1]
fences <- c(Lower = unname(q[1] - 1.5 * iqr), Upper = unname(q[2] + 1.5 * iqr))
flagged <- cost < fences[1] | cost > fences[2]
data.frame(Lower_fence = fences[1], Upper_fence = fences[2],
           Flagged_products = sum(flagged), row.names = NULL)
```

```
  Lower_fence Upper_fence Flagged_products
1       0.025       0.305                0
```

- No cost is outside these fences.
- That does not prove the data are error-free or normally distributed.
- A boxplot alone also cannot establish symmetry; inspect the histogram and the recorded values.

Your turn

- Compare the histogram and boxplot.
- What can you see more easily in each?
- Explain why a legitimate high price should not be deleted simply to make a distribution look more balanced.

7. Write the business interpretation

Here is one way to report the results to the buyer:

> - In this historical list of 33 cereal products, both the mean and median cost are about $0.17 per ounce.
> - Individual costs range from $0.10 to $0.27, with a sample standard deviation of about $0.047 per ounce.
> - The middle half lies between $0.13 and $0.20.
> - A single typical cost is therefore useful as an overview, but it hides meaningful differences between products.

- These results describe the listed products.
- They do not establish current market prices, explain why products have different costs, or show which product the shop should stock.
- A stocking decision would also need information about customer demand, selling prices, margins and other product characteristics.

8. Your turn: a lower-cost shortlist

The buyer wants to examine products costing **no more than $0.15 per ounce in this historical list**.

- Use the starter code to create the shortlist.
- Then apply the functions used above.

```
# subset() keeps rows meeting the condition; <= includes exactly $0.15.
lower_cost <- subset(cereals, cost_per_ounce <= 0.15)

# Complete your analysis below.
# 1. How many products are included?
# 2. What are their mean, median, sample SD and IQR of cost per ounce?
# 3. How do these compare with the full list?
```

Submit your completed notebook with:

- A histogram or boxplot.
- A summary table containing mean, median, sample SD and IQR for both lists.
- **Three or four sentences for the buyer**.
- Explain whether the shortlist has a lower typical cost and less variation.
- State why selecting a low-cost subset does not, by itself, identify a better product or demonstrate a price reduction over time.
- As a small extension, change the histogram's `binwidth` from `0.02` to `0.04`.
- Describe what changes in the picture and what remains unchanged in the calculated summaries.

Source and data notes

- Dataset: **CEREALS**, supplied with *Basic Business Statistics: Concepts and Applications*, ninth edition, Berenson, Levine and Krehbiel; see exercise 3.68, pp. 127–128.
- The book credits Consumers Union of U.S., Inc., *Consumer Reports*, October 1999, pp. 33–34.
- That magazine attribution is reported from the book and has not been independently checked against the original article.
- The course CSV preserves all 33 rows, numerical values and product labels from the publisher's `Data` worksheet.
- Column names were simplified for R.
- Data notes describe the fields and limitations.

Continue with Practical 3: Association and group comparisons.